

SEVERO-YENISEYSKIY, Russian Federation

WMO: 239860

Lat: 60.3736N

Lon: 93.0172E

Elev: 511

StdP: 95.33

Time Zone: 7.00 (E07)

Period: 09-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a) 1	(b) -39.2	(c) -36.3	(d) -42.5	(e) 0.1	(f) -39.3	(g) -39.5	(h) 0.1	(i) -36.2	(j) 7.9	(k) -15.2	(l) 7.2	(m) -15.2	(n) 0.8	(o) 250	(p) N/A

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a) 7	(b) 9.3	(c) 27.6	(d) 16.3	(e) 25.6	(f) 15.5	(g) 23.7	(h) 15.0	(i) 18.0	(j) 23.6	(k) 17.1	(l) 22.9	(m) 16.2	(n) 21.7	(o) 2.6	(p) 200

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
16.1	12.2	19.7	15.0	11.3	19.2	13.9	10.5	18.2	53.0	23.6	50.1	23.1	47.3	21.8	21.1

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a) 7.2	(b) 6.2	(c) 5.4	(e) -40.0	(f) 30.1	(g) 4.3	(h) 1.9	(i) -43.2	(j) 31.4	(k) -45.7	(l) 32.5	(m) -48.1	(n) 33.6	(o) -51.3	(p) 35.0		
(a) 7.2	(b) 6.2	(c) 5.4	(e) -40.1	(f) 20.0	(g) 4.3	(h) 0.8	(i) -43.2	(j) 20.6	(k) -45.7	(l) 21.0	(m) -48.2	(n) 21.4	(o) -51.3	(p) 22.0		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6)	Temperatures, Degree-Days and Degree-Hours	DBAvg	-2.9	-22.2	-18.6	-9.4	-1.4	3.9	15.0	16.5	13.0	5.7	-2.9	-15.7	-19.5	(6)
(7)		DBStd	14.79	7.38	6.91	6.24	5.13	5.45	5.17	3.89	3.89	4.95	6.12	7.93	8.96	(7)
(8)		HDD10.0	5219	997	799	603	343	206	16	3	15	149	401	772	914	(8)
(9)		HDD18.3	7796	1256	1033	861	593	448	122	81	171	380	658	1022	1172	(9)
(10)		CDD10.0	513	0	0	0	0	17	166	204	107	19	0	0	0	(10)
(11)		CDD18.3	51	0	0	0	0	0	22	23	5	0	0	0	0	(11)
(12)		CDH23.3	481	0	0	0	0	0	242	197	40	2	0	0	0	(12)
(13)		CDH26.7	89	0	0	0	0	0	49	33	6	0	0	0	0	(13)
(14)	Wind	WSAvg	2.5	2.2	2.3	2.9	3.2	3.0	2.2	2.1	2.2	2.4	2.6	2.5	2.2	(14)
(15)	Precipitation	PrecAvg	658	33	32	36	42	49	66	70	82	72	75	54	46	(15)
(16)		PrecMax	1297	83	67	93	107	162	193	221	229	188	153	141	113	(16)
(17)		PrecMin	367	5	10	6	6	10	14	7	23	5	29	24	15	(17)
(18)		PrecStd	229	18	14	19	22	31	37	51	45	43	29	26	22	(18)
(19)	Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	-6.2	-3.1	6.3	13.8	21.4	30.2	29.6	27.5	22.4	12.2	0.8	-2.9	(19)
(20)			MCWB	-6.9	-3.9	1.6	5.5	10.4	17.3	18.3	17.0	13.4	7.4	0.2	-3.3	(20)
(21)		2%	DB	-8.2	-4.8	3.4	10.1	19.0	28.0	27.4	24.0	18.7	9.5	-0.3	-4.6	(21)
(22)			MCWB	-8.7	-5.5	-0.8	4.0	10.4	16.0	16.5	16.2	11.5	6.2	-0.8	-5.1	(22)
(23)		5%	DB	-10.0	-7.0	0.7	8.1	16.2	26.2	25.6	22.0	15.7	7.2	-1.6	-6.0	(23)
(24)			MCWB	-10.5	-7.7	-1.8	3.0	9.2	15.2	16.0	15.2	10.6	5.0	-2.1	-6.5	(24)
(25)		10%	DB	-11.5	-9.2	-1.1	5.8	13.0	24.2	23.7	19.6	13.1	5.2	-3.8	-7.9	(25)
(26)			MCWB	-11.9	-10.0	-3.0	1.6	7.4	14.4	15.3	14.2	9.0	3.6	-4.4	-8.4	(26)
(27)	Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	-6.5	-3.7	2.3	7.1	12.8	18.9	19.7	18.9	14.3	8.4	0.5	-3.4	(27)
(28)			MCDWB	-6.1	-3.3	6.1	11.9	17.3	26.2	26.7	23.8	19.7	10.5	0.7	-3.1	(28)
(29)		2%	WB	-8.7	-5.6	-0.3	4.9	11.0	17.4	18.2	17.1	12.6	6.7	-0.8	-5.0	(29)
(30)			MCDWB	-8.2	-4.9	2.3	9.2	17.5	24.7	24.0	22.4	17.9	8.6	-0.4	-4.6	(30)
(31)		5%	WB	-10.4	-7.6	-1.6	3.4	9.6	16.5	17.4	16.1	10.9	5.2	-2.0	-6.4	(31)
(32)			MCDWB	-10.0	-7.0	0.5	7.0	15.4	24.0	22.7	20.7	14.1	7.0	-1.5	-6.0	(32)
(33)		10%	WB	-12.0	-9.9	-2.8	2.3	7.6	15.4	16.5	14.9	9.6	3.5	-4.4	-8.4	(33)
(34)			MCDWB	-11.5	-9.3	-1.1	5.2	12.8	22.7	21.5	18.8	13.0	5.2	-3.9	-7.8	(34)
(35)	Mean Daily Temperature Range	5% DB	MDBR	6.2	7.2	8.3	7.9	8.3	10.2	9.3	8.6	6.7	5.0	6.3	5.9	(35)
(36)			MCDBR	6.9	6.3	8.6	9.8	11.8	13.5	12.4	11.7	10.5	7.2	5.2	5.5	(36)
(37)			MCWBR	6.7	6.0	6.2	6.0	6.3	5.4	5.0	5.3	5.3	5.1	5.2	5.3	(37)
(38)		5% WB	MCDBR	7.1	6.5	7.9	8.6	10.9	11.6	9.7	10.5	9.3	6.6	5.8	5.5	(38)
(39)			MCWBR	6.9	6.2	6.1	5.8	6.3	5.3	4.8	5.3	5.3	4.9	5.8	5.4	(39)
(40)	Clear-Sky Solar Irradiance	taub	0.246	0.282	0.312	0.342	0.358	0.371	0.415	0.384	0.332	0.299	0.242	0.217	(40)	
(41)		taud	1.972	2.032	2.052	2.090	2.308	2.402	2.263	2.328	2.449	2.300	2.021	1.901	(41)	
(42)		Ebn at Noon	480	687	796	845	865	859	803	801	793	683	482	348	(42)	
(43)		Edh at Noon	47	81	111	130	114	106	119	103	77	62	43	32	(43)	
(44)	All-Sky Solar Radiation	RadAvg	0.43	1.36	2.71	4.21	4.91	5.46	4.97	3.62	2.18	0.98	0.47	0.22	(44)	
(45)		RadStd	0.06	0.11	0.25	0.36	0.42	0.67	0.43	0.28	0.35	0.15	0.07	0.04	(45)	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)		
(46) Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (4 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page